Unit 1:

Information Systems



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Learning Outcomes

- **After successfully completing this unit, the learner will:**
  1. Understand the capabilities of different types of information system
  2. Understand the roles of information systems in organisations
  3. Understand development concepts relevant to information systems
  4. Understand professional issues relevant to information systems

Learning Outcome 1:

Understand the capabilities of different types of information system

Introduction

Definitions of

1. Systems
2. Information Systems
3. Subsystems

Systems

- A system is a collection of interrelated components that function together to achieve some outcome .

Information Systems

- An information system is a collection of interrelated components that collect, process, store, and provide as output the information needed to complete the business task.

Subsystems

- A subsystem is a system that is part of another system, so subsystems might be one way to think about the components of a system.

Communications computer package used to support a given activity

1. Wireless Messaging Services
2. Wireless internet access point.
3. Cybercafes
4. Global positioning systems.

Wireless Messaging Services

- Users can send and receive wireless messages to and from smart phones, cellphones, handled game consoles and other mobile devices eg. Text messaging, wireless instant messaging.

Wireless internet access point.

- At home, work, school, and in many public locations, people connect wirelessly to the internet through a wireless internet access point using mobile computers, smart phones etc.

Cybercafes

- A cybercafe or internet café is a coffeehouse, restaurant, or other location that provides personal computers with internet access to its customer for communication.

Global positioning systems.

- A global positioning system (GPS) is a navigation system that consists of one or more earth-based receivers that accept and analyse signals sent by satellites in order to determine the receivers geographic location.

Features and capabilities of different types of information systems

1. A customer Relationship Management system
2. A supply chain management system
3. Accounting and financial management system
4. A manufacturer management system

A customer Relationship Management System

- A customer relationship management system is a system that supports marketing, sales and service operations involving direct and indirect customer interactions.

A supply chain management system

- A supply chain management system is a system that seamlessly integrates product development, product acquisition, manufacturing and inventory management

Accounting and financial management system

- Accounting and financial management system is a system that records accounting information needed to produce financial statements and other reports such employee-related tasks as payroll, benefits, hiring and training.

A manufacturer management system

- A manufacturer management system is a system that controls internal production processes that turn raw materials into finish goods.

Organisational needs and systems

- An organization system is the structure of how an organization is set up.
- A good organizational structure lays out both a hierarchy and the flow of communication in a company.
- It is important for every business, no matter its size, to implement an organizational systems.

Databases or systems that support data intensive applications

- **Database** is a collection of data organized in a manner that allows access, retrieval, and use of that data.

Types of Databases or systems that support data intensive applications

- Three popular data models in use today are
    1. Relational,
    2. Object-oriented,
    3. Multidimensional.

Relational Database

- A relational database is a database that stores data in tables that consist of rows and columns.
- In addition to storing data, a relational database also stores data relationships.
- A relationship is a link within the data.
- Applications best suited for relational databases are those whose data can be organized into a two-dimensional table, that is, tables with rows and columns.
- Many organizations use relational databases for payroll, accounts receivable, accounts payable, general ledger, inventory, order entry, invoicing, and other business- related functions.

Object-Oriented Database

- An *object-oriented database* (*OODB*) stores data in objects.
- An *object* is an item that contains data, as well as the actions that read or process the data.
- Examples of applications appropriate for an object-oriented database include media databases that store images, audio clips, and/or video clips; groupware databases that store documents, such as schedules, calendars, manuals, memos, and reports; and CAD (computer-aided design) databases that store data about engineering, architectural, and scientific designs.

Multidimensional Database

- A multidimensional database stores data in dimensions.
- Whereas a relational database is a two-dimensional table, a multidimensional database can store more than two dimensions of data.
- These multiple dimensions allow users to access and analyze any view of the database data.
- One application that uses multidimensional databases is a data warehouse.
- A data warehouse is a huge database that stores and manages the data required to analyze historical and current transactions

Practices used to increase efficiency and effectiveness of information systems and business practice

1. Systems and innovation opportunities
2. Systems and customer delight
3. Systems and organizational productivity
4. Systems and workers productivity
5. Information systems as a value add for organization

Systems and innovation opportunities

- Development in information systems has brought opportunities but also threats.
- The onus is on the organization to identify opportunity and implement it.
- Organisation needs to develop strategies which can best utilize information systems to increase overall productivity.

Systems and customer delight

- Organisation are fully aware that proliferation of information systems has reduced product life cycle, reduced margin and brought in new products.
- In such scenario customer satisfaction alone will not suffice, organization needs to strive for customer delight.

Systems and organizational productivity

- Organisations require different types of information systems to mitigate distinctive process and requirements.
- Efficient business transaction systems makes organization productive.
- Business transaction systems ensure that routine process are captured and acted upon effectively, for example, sales transaction, cash transaction, payroll etc.

Systems and workers productivity

- Information systems have facilitated the increase in workers productivity.
- With introduction of email, video conferencing and shared white board collaboration across organisation and department have increase.

Information systems as a value add for organization

- Organisation use information systems to achieve its various strategy as well as short-term and long-term goals.
- Department of information systems was to improve productivity and business effectiveness of organisation.

Common examples of information systems used commercially including

1. Systems to support the travel industry,
2. Airport check-ins,
3. Banking,
4. Payrolls,
5. Management information and recognising the important characteristics of information systems being flexible and maintainable .

The different types of data guarantees given by an information system, such as support for transactions

1. **Transaction processing systems (TPS)** are basic business systems that serve the operational level of the organization by recording the daily routine transactions required to conduct business, such as payroll and sales receipts.

2. **Management information systems (MIS)** serve middle managers' interests by providing current and historical performance information to aid in planning, controlling, and decision making at the management level. MIS typically compress TPS data to present regular reports on the company's basic operations.

3. **Decision support systems (DSS)**, or business intelligence systems, help managers with non-routine decisions that are unique, rapidly changing, and not easily specified in advance. DSS are more analytical than MIS, using a variety of models to analyze internal and external data or condense large amounts of data for analysis.

4. **Executive support systems (ESS)** provide a generalized computing and communications environment that help senior managers address strategic issues and identify long-term trends in the firm and its environment. ESS address nonroutine decisions requiring judgment, evaluation, and insight because there is no agreed-on procedure for arriving at a solution. ESS present graphs and data from many internal and external sources through an interface that is easy for senior managers to use. Often the information is delivered to senior executives through a **portal**, which uses a Web interface to present integrated personalized business content.

Methods of interrogating information systems for information and producing report

1. Review of Records, Procedures, and Forms
2. Observation
3. Joint Application Development (JAD)
4. Interviewing

Review of Records, Procedures, and Forms

- Review of existing records, procedures, and forms helps to seek insight into a system which describes the current system capabilities, its operations, or activities.

Advantages

1. It can provide a clear understanding about the transactions that are handled in the organization, identifying input for processing, and evaluating performance.
2. It can help an analyst to understand the system in terms of the operations that must be supported.

Observation

- This is a method of gathering information by noticing and observing the people, events, and objects.
- The analyst visits the organization to observe the working of current system and understands the requirements of the system.

Advantages

1. It is a direct method for gleaning information.
2. It is useful in situation where authenticity of data collected is in question or when complexity of certain aspects of system prevents clear explanation by end-users.
3. It produces more accurate and reliable data.
4. It produces all the aspect of documentation that are incomplete and outdated

Joint Application Development (JAD)

- It is a new technique developed by IBM which brings owners, users, analysts, designers, and builders to define and design the system using organized and intensive workshops.
- JAD trained analyst act as facilitator for workshop who has some specialized skills.

Advantages of JAD

1. It saves time and cost by replacing months of traditional interviews and follow-up meetings.
2. It is useful in organizational culture which supports joint problem solving.
3. Fosters formal relationships among multiple levels of employees.
4. It can lead to development of design creatively.
5. It Allows rapid development and improves ownership of information system.

Interviewing

- Systems analyst collects information from individuals or groups by interviewing.
- The analyst can be formal, legalistic, play politics, or be informal; as the success of an interview depends on the skill of analyst as interviewer.

Advantages of Interviewing

1. This method is frequently the best source of gathering qualitative information.
2. It is useful for them, who do not communicate effectively in writing or who may not have the time to complete questionnaire.
3. Information can easily be validated and cross checked immediately.
4. It can handle the complex subjects.
5. It is easy to discover key problem by seeking opinions.
6. It bridges the gaps in the areas of misunderstandings and minimizes future problems.

What is office automation systems

- Office automation systems refers to the varied computer machinery and software used to digitally create, collect, store, manipulate, and relay office information needed for accomplishing basic tasks.

How common office applications may be automated to create information systems

1. Office Automation Systems (OAS) as configurations of networked computer hardware and software
2. Combination of tools available to create, store, manipulate and relay office information over a network
3. OAS systems applied to business and communication functions previously performed manually
4. Types of functions include electronic publishing (word-processing, advanced copying, scanning and printing processes), electronic communication, need for integrated communication that is higher and more complex than before, system to map, transform and transfer data of large volume between different information systems (such as JDBC or ODBC connections, web services)
5. Understanding the difference between online (synchronous) and offline (asynchronous) processing, batch processing

- Electronic collaboration (e.g. groupware software enabling simultaneous working on same file), image processing (for forms, medical records, dental records or mortgage applications) and office management (organising appointments, projects and data)
- Integration of front office and back office systems and processes
- Managing and tracking orders, processing of products and services including offer and pricing through a variety of media such as catalogues, advertisements and websites

Integration of front office and back office systems and processes

- Front office operations systems handle prospect and client related activities such as marketing, quotation, order management, and customer service.
- Front office processes are much different. Because of the unpredictable nature of customer activities, front office operational systems pose more control challenges.

- Back office operations on the other hand, address order fulfilment and accounting activities.
- Back office processes deal with knowable types of activities because there is higher level of internal control over such processes.

Office Automation Systems (OAS) as configurations of networked computer hardware and software

- A variety of office automation systems are now applied to business and communication functions that used to be performed manually or in multiple locations of an organisation, such as preparing written communications and strategic planning.

Types of functions integrated by office automation systems include

1. Electronic publishing
2. Electronic communication
3. Electronic collaboration
4. Image processing
5. Office management.

Electronic publishing (word-processing, advanced copying and scanning and printing processes)

- Electronic publishing systems include word processing and desktop publishing.
- Word processing software e.g. Microsoft Word, Corel Word-Perfect allows users to create, edit, revise, store and print documents such as letters, memo, report, and manuscripts.

Electronic communication

- Electronic communication systems include electronic mail (email), voice mail, facsimile (fax), and desktop videoconferencing.

Electronic collaboration

- Electronic collaboration is made possible through electronic meeting and collaborative work systems and teleconferencing.
- Electronic meeting and collaborative work systems allow teams of coworkers to use networks of microcomputers to share information, update schedules and plans, and cooperate on projects regardless of geographic distance.

Image processing

- Image processing systems include electronic document management, presentation graphics, and multimedia systems.
- Imaging systems into digital form that can be stored in a computer system.

Office management.

- Office management systems include electronic office accessories, electronic scheduling, and task management.
- These systems provide an electronic means of organizing people, projects, and data,

Combination of tools available to create, store, manipulate and relay office information over a network

1. Data storage and manipulation
2. Data management
3. Data exchange
4. Business software

Data storage and manipulation

- Data storage includes important and confidential office documents and records.
- The office automation system includes data application, which are used to create or edit document, file, spreadsheet, or image.

Data management

- Data management is one of the major components of an office automation system that offers strategic advantages by simplifying the management of stored data and information.

Business software

- Other tools that are specific to a department or industry such as management, marketing, sales and technology management platforms.

Data exchange

- Exchange of stored or manipulated data and information is also an important tool of an office automation system.
- Sending files or exchanging data or information between one or more than one member of an organisation has become possible with an electronic transfer application.

OAS systems applied to business and communication functions previously performed manually

1. Storage of information
2. Data exchange
3. Data management

Types of functions include

1. Electronic publishing (word-processing, advanced copying and scanning and printing processes)
2. Electronic communication
3. Need for integrated communication that is higher and more complex than before,
4. System to map,
5. Transform and transfer data of large volume between different information systems (such as JDBC or ODBC connections, web services)

Electronic publishing (word-processing, advanced copying and scanning and printing processes)

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Electronic communication

- Electronic communication systems include electronic mail (email), voice mail, facsimile (fax), and desktop videoconferencing.

Need for integrated communication that is higher and more complex than before

- The purpose of integrated communication that is higher and more that before is an approach to promoting a message through multiple strategies that work together and reinforce one another.
- For example, company my promote a new logo, slogan or strategy through multiple media such as print, television, web and social networks.

System to map

- System to map shows the components and boundary of a system and the components of the environment at a point in time.
- The main use of a system to map is to help structure a system and communicate the results to others.

Transform and transfer data of large volume between different information systems such as

1. JDBC or ODBC connections,
2. Web services

JDBC or ODBC connections

- **JDBC** stands for Java Database Connectivity is an application interface for the programming language java, which defines how a client may access a database.
- **ODBC** stands for Open Database Connectivity is a standard application programming interface (API) for accessing database management systems

Advantages of ODBC

1. ODBC provides a consistent interface regardless of the kind of database server used.
2. You can have more than concurrent connection.
3. Applications do not have to be bound to each database on which they will run.
4. It helps to create a new IGSS object buy inserting the record of the drawing in the table.
5. Helps to edit multiple objects at a time by the help of template provided
6. Can easily access configuration data with the use of visual basic.

How ODBC works

- ODBC consists of four components, working together to enable functions or uses
  1. Application
  2. Driver manager
  3. Driver
  4. Data source

Application

- Processes and call the ODBC functions and submits the SQL statements.

Driver manager

- Loads drivers for each application

Driver

- Handles ODBC function calls, and then submits each SQL request to a data source;

Data source

- The data being accessed and its database management operating system.

Web services

- A web service is any piece of software that makes itself available over the internet and uses a standardized XML message systems.

Understanding the difference between online (synchronous) and offline (asynchronous) processing, batch processing

Online (synchronous) and offline (asynchronous) processing

- **Online (synchronous) processing** is online or distance education that happens in real time, often with a set class schedule and required login times.
- **Offline (asynchronous) processing** is an offline or distance education that does not require real-time interaction;

Batch processing

- Batch processing is a technique in which an operating system collects the programs and data together in a batch before processing starts.

Advantages

- Batch processing takes much of the work of the operator to the computer.
- Increased performance as a new job get started as soon as the previous job is finished, without any manual intervention.

Disadvantages

- Difficult to debug program.
- A job could enter an infinite loop.
- Due to lack of protection scheme, one batch job can affect pending jobs.

Electronic collaboration

- Electronic collaboration is made possible through electronic meeting and collaborative work systems and teleconferencing.
- Electronic meeting and collaborative work systems allow teams of coworkers to use networks of microcomputers to share information, update schedules and plans, and cooperate on projects regardless of geographic distance.

Examples of Electronic collaboration

1. Groupware software enabling simultaneous working on same file
2. Image processing (for forms, medical records, dental records or mortgage applications) and
3. Office management (organising appointments, projects and data)